CAPSULE-CARE

Smart Queue & Medicine Reminder System

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MCA Department

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1. Abstract

Capsule-Care is a web-based smart queue and medicine reminder system designed for clinics and small hospitals. Patients can book appointments, receive token numbers, and track their queue position in real time. Doctors can manage the queue, attend patients one by one, mark them as paid or completed, and download daily reports. The system also provides medicine dosage reminders after consultation. This eliminates long physical waiting lines and helps both doctors and patients save time.

2. Problem Statement

In most clinics and hospitals, patients wait in long physical queues without knowing their turn. Doctors have no digital way to manage patient flow, track payments, or send medicine reminders. This leads to confusion, delays, and a poor experience for both patients and doctors. There is no simple, affordable system that combines queue management with medicine reminders.

3. Proposed Solution

Capsule-Care solves this by providing a token-based smart queue system where patients book online, get a token number, and wait from home. Doctors start the queue, attend patients one by one, and mark them done. Paid patients get queue priority. After consultation, patients receive medicine reminders. The entire system runs on a web browser — no app installation needed.

4. System Architecture

The system follows a 3-tier web architecture with a Flask backend, MySQL database, and HTML/CSS frontend.

Presentation Layer (Frontend)

HTML5 + CSS3 + Jinja2 Templates
Responsive UI with Dark/Light Theme
Token Card (PDF Download)
Real-time Queue Status Board

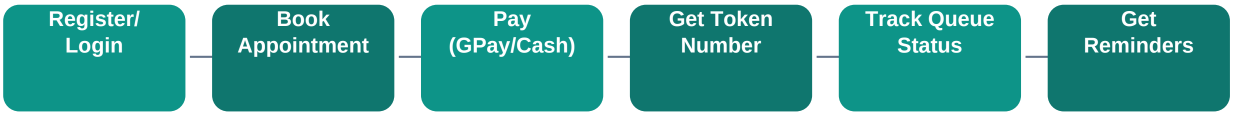
Application Layer (Backend)

Python Flask Web Framework
Session-based Auth (Patient + Doctor)
Queue Engine with Token Priority Logic
Payment Tracking (GPay / Cash)

Data Layer (Database)

MySQL Database (Aiven Cloud hosted)
Tables: patients, doctors, appointments, reminders, queue_sessions
Auto-initialize with DDL on first run

5. Patient Flow



5.1 Register / Login — Patient creates an account with name, age, email, mobile and password. Login with email and password.

5.2 Book Appointment — Select a doctor, date and time slot. System assigns a token number automatically. If GPay is chosen, payment is confirmed later on the token page. Cash is paid at clinic.

5.3 Queue Tracking — Live queue board shows all tokens with their status (Waiting, Attending, Done). Paid patients get higher priority in the queue.

5.4 Token Card — View and download a PDF token card with QR code, doctor name, date and token number.

5.5 Medicine Reminders — After consultation, patients can set medicine reminders with dosage, frequency and food timing.

6. Doctor Flow

6.1 Register / Login — Doctor creates profile with name, specialization, fees and optional GPay QR code.

6.2 Start Queue — Doctor clicks 'Start Queue' button to begin attending. Until then all patients see 'Waiting'.

6.3 Mark Paid / Unpaid — Doctor can toggle payment status. Paid tokens jump ahead in queue priority.

6.4 Mark Done — Doctor marks a token as completed. Patient's token becomes inactive.

6.5 Postpone / Cancel — Doctor can postpone the whole day's queue to a future date or cancel individual appointments.

7. Key Features

Smart Queue Management — Token-based queue with auto-numbering. Paid patients get priority. Real-time status board.

Postpone & Cancel — Doctor can postpone the whole day's queue. Patients see red alert with new date. Cancel Postpone restores waiting.

Payment Tracking — GPay or Cash payment. Doctor can toggle Paid/Unpaid. Paid tokens move ahead in queue.

Token PDF Download — Downloadable appointment token card with QR code, doctor details and token number.

Medicine Reminders — Set dosage, frequency, food timing and duration. Browser notification support.

Doctor Reports — Daily PDF report of completed tokens with patient details.

Dark / Light Theme — Toggle between themes. Saved in browser preference.

Responsive Design — Works on mobile, tablet and desktop browsers.

8. Database Schema

Table	Key Columns	Purpose
patients	id, name, age, email, mobile, password	Patient accounts and login
doctors	id, doctor_name, username, email, fees, gpay_qr	Doctor profiles and payment QR
appointments	id, patient_id, doctor_id, token_number, status, payment_status, postponed_date	All booking and queue state
reminders	id, patient_id, medicine_name, dosage, food_timing	Medicine dosage reminders
queue_sessions	doctor_id, queue_date, started_at	Tracks when queue was started

Appointment statuses: waiting, done, postponed, absent, cancelled

Payment statuses: pending, paid, cash

9. Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, Jinja2 Templates, SVG Icons
Backend	Python 3.13, Flask 3.0.3
Database	MySQL 8.0 (Aiven Cloud free tier)
PDF Generation	ReportLab 4.2.2
Deployment	Vercel (Serverless Functions)
Version Control	Git + GitHub
Auth	Session-based with Werkzeug password hashing

10. Deployment Steps

1. Push source code to GitHub repository.
2. Connect GitHub repo to Vercel (vercel.com → New Project → Import).
3. Set Framework to 'Other', Root Directory to './'.
4. Add Environment Variables: DB_HOST, DB_PORT, DB_USER, DB_PASSWORD, DB_NAME, SECRET_KEY.
5. Deploy and wait for build to complete.
6. Visit /setup endpoint once to create database tables.
7. (Optional) Set up UptimeRobot to ping /health every 5 minutes to keep Aiven DB awake.

11. Project Structure

```
hackathon/

app.py — Main Flask application (all routes, DB, auth, PDF)

schema.sql — Database DDL (backup reference)

requirements.txt — Python dependencies

vercel.json — Vercel deployment config

api/index.py — Vercel serverless entry point

static/css/style.css — All styles (light + dark theme)

templates/ — 18 Jinja2 HTML templates
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12. Future Enhancements

- SMS / Email appointment confirmations
- Doctor availability calendar

• Patient review and rating system
• Multi-language support (Tamil + English)
• Admin dashboard for hospital management

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